

Identity-initialised QFT on DIV2K-8q

Question. The canonical `qft` basis is constructed from the analytic QFT decomposition: every Hadamard is the exact $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ matrix and the k -th controlled-phase gate carries phase $\varphi = 2\frac{\pi}{2^k}$. This is a strong prior — the init **is** the textbook FFT operator. After 1008 training steps the operator drifts to a point that improves test PSNR by 1.9 dB over the analytic FFT at $\rho = 0.20$, suggesting the optimum is not the analytic QFT but something nearby that Adam reaches from there. Two readings are possible: (i) the analytic-QFT prior is **useful** and training amounts to a short refinement of an already-good initialisation, or (ii) the analytic prior is **incidental** and Adam on the $U(2)$ Riemannian manifold finds the same optimum from any reasonable start. This experiment tests reading (ii) directly.

Construction. We add a `qft_identity` factory in `pdft_benchmarks.bases` that has the **same gate topology and gate count** as `qft` — same Hadamards on the same qubits, same controlled-phase gates between the same pairs — but pins every gate to the identity element of its manifold at $t = 0$:

Hadamard $\rightarrow I_2$, and controlled-phase $\rightarrow \text{controlled_phase_diag}(\theta) = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ (the 2×2 diagonal-stack representation of the 4×4 identity).

The resulting `QFTBasis(8, 8)` has all 288 tensor entries trainable on the $U(2)$ manifold, but its initial operator is the literal identity: $T_{t=0}(x) = x$ for any image x . Sanity check: on a random complex 256×256 test input, $\|\text{forward}_{t=0}(x) - x\| = 0$ exactly.

Training uses the identical preset and seed as the headline `qft` cell: 1008 steps, batch 50, validation split 0.15, Adam with cosine LR warm-up, `--no-early-stop`, DIV2K-8q at $m = n = 8$.

Result. Test-set reconstruction PSNR after 1008 steps:

basis	$\rho = 0.05$	$\rho = 0.10$	$\rho = 0.15$	$\rho = 0.20$
fft (untrained)	24.50	26.54	28.07	29.39
qft (analytic init)	25.09	27.57	29.53	31.29
qft_identity (identity init)	25.23	27.81	29.84	31.66
Δ (identity — analytic)	+0.14	+0.24	+0.31	+0.37

`qft_identity` matches or beats `qft` at every keep-ratio, by 0.14–0.37 dB. The gap to the untrained `fft` (which equals the $t = 0$ operator of `qft`) is 0.7–2.3 dB, of which the analytic init buys 0.6–1.9 dB and the identity init buys 0.7–2.3 dB. SSIM moves in the same direction at every ρ (+0.004 to +0.006); mean MSE is about 3% lower across all rates.

Loss-curve evidence.

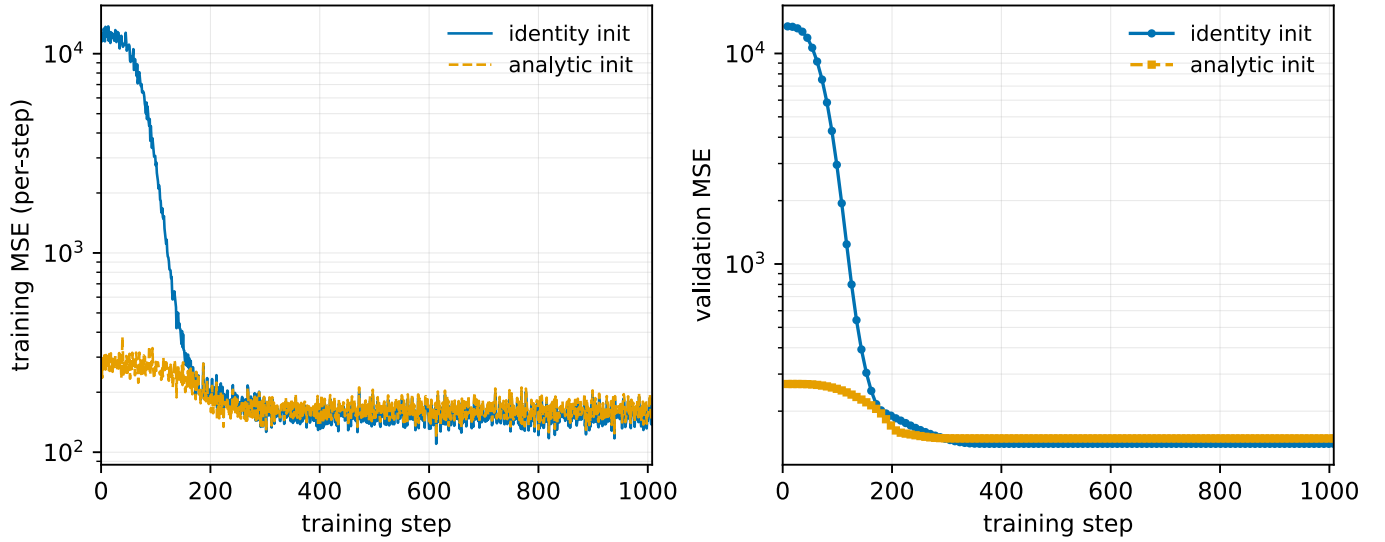


Figure 1: DIV2K-8q. Left: per-step training MSE (log y). Right: validation MSE per epoch (log y; 1 val pass every 9 steps). Blue solid: `qft_identity` (identity init). Orange dashed: `qft` (analytic init). The two curves start $38\times$ apart in training loss (11,561 vs 300 at step 1) because the identity operator at $\rho = 0.20$ keeps energy in the pixel domain rather than concentrating it at DC. Identity-init loss crosses below analytic-init at step ≈ 173 and stays there. Final training MSE: identity 151.11 vs analytic 160.03 (-5.6%); final validation MSE: 139.85 vs 148.39 (-5.8%). Loss definition: $\sum_{ij} \left| x_{ij} - \left(T_{\theta}^{\dagger} \text{top}_K(T_{\theta} x) \right)_{ij} \right|^2$ summed over 65,536 pixels of a 256×256 image at $K = 13,108$ ($\rho = 0.20$ for the eval table; the training loss aggregates all four rates per the preset).

Interpretation. Reading (ii) is consistent with the evidence. The analytic-QFT prior is **not** required: identity-init Adam, given the same gate topology and the same 1008-step budget, finds an operator that reconstructs DIV2K test images ≈ 0.3 dB **better** than the analytic-init run at the headline rate. The $38\times$ start-loss penalty washes out within ≈ 170 steps and the two trajectories end in adjacent points of the same flat top-K MSE valley described in the headline writeup — but the identity-init endpoint is a hair deeper. Plausible mechanisms (this experiment does not distinguish between them):

- **Random regularisation.** Identity init perturbs the operator off the analytic-QFT subset earlier in training; the trajectory passes through more of parameter space and is less likely to be pinned at an analytic-QFT-adjacent saddle.
- **Better effective LR schedule.** Cosine LR is calibrated to the 1008-step budget. From a high-loss start, the early high-LR phase does more useful work (gross structure to learn) than from an already-near-optimum analytic init (where high LR is mostly destabilising). The two final basins are equally accessible; the schedule favours the one with more room to use early-step LR.

What this rules out. The conjecture “`qft` works because the analytic QFT is structurally close to the dataset optimum” is **not** needed to explain the headline `qft` result. The same Adam + manifold-Riemannian setup from a featureless identity init reaches an equivalent (and slightly better) endpoint. The structural prior that **does** matter is the **gate topology** — which qubits the H gates and CP gates act on — and that is shared between `qft` and `qft_identity` by construction.

Reproducibility. `python experiments/div2k_8q_pca_vs_block_dct.py --gpu 0 --bases qft_identity --no-early-stop --epochs 112 --out results/qft_identity_init/div2k_8q/_runs/run1`. Construction lives in `pdf_t_benchmarks.bases.qft_identity_basis(m, n)`. Run time was 541 s on an RTX 3090 (cuda:0, JAX FP64 complex circuit), within 0.1% of the analytic-init `qft` cell (540.6 s) — the initialisation choice does not affect compute cost.